Hospital Management System

Executive summary

For an MVP go with Fast2SMS + Amazon SES + Razorpay + DigitalOcean + Firebase + Sentry (lowest monthly cost + reasonable reliability). For a production hospital-grade setup choose Twilio + SendGrid or Amazon SES + Razorpay + AWS (EC2 + RDS + S3) + Sentry/Datadog (best reliability, global SLAs, advanced tooling).

Key citations for pricing/regulatory notes: Twilio SMS pricing, Fast2SMS pricing, MSG91 pricing, SendGrid pricing, Razorpay pricing, DLT registration details**.**

Vendor selection table

Each category: Best (enterprise) / Cheapest (MVP) — with why, typical cost, key pros/cons, onboarding notes.

1) SMS & OTP (transactional) — must have DLT registration

- **Best (enterprise): Twilio (Global + India)**
    - Why: enterprise SLA, global reach, advanced APIs (verify, programmable SMS), high deliverability & monitoring.
    - Typical cost signal: international pricing pages show per-message costs and enterprise support options.
    - Pros: great SDKs, fallback routing, phone number provisioning, detailed delivery reporting.
    - Cons: more expensive than local aggregators; sender ID / DLT templates still required.
    - Onboard notes: sign up, apply for Alphanumeric Sender ID via Twilio (or use aggregator to register DLT headers).
- **Cheapest (MVP): Fast2SMS (or MSG91)**
    - Why: very low per-SMS rates in India for transactional OTP and reminders; easy to buy prepaid packs.
    - Typical cost (India): ₹0.11 – ₹0.25 per SMS at scale (can start with small prepaid balance).
    - Pros: cheap, fast onboarding, local DLT experience.
    - Cons: lower SLA guarantees, single-region dependability; some advanced routing features missing.
    - Onboard notes: you must do DLT registration (Entity + Sender ID + Templates) — see DLT guides and expected one-time/annual costs (≈ ₹5,900 entity + ₹590 per sender ID in many operators).

DLT (must) — register your principal entity, register sender ID(s), register message templates and consent templates. Use aggregator help if you want them to do DLT for you (they'll charge a setup fee).

2) Email (transactional receipts, appointment confirmations)

- **Best (enterprise): SendGrid**
  - Why: easy developer UX, deliverability tooling, templating & analytics. Pricing tiers and support for high-volume transactional emails.
  - Pros: great UI, deliverability dashboards, templates, SDKs, webhooks.
  - Cons: costs rise with volume.
- **Cheapest (MVP): Amazon SES (via AWS)**
  - Why: extremely low cost per email, rock-solid deliverability when set up properly (SPF/DKIM).
  - Typical cost: fractions of a cent per email (very cheap compared to others).
  - Onboard notes: configure verified domain, SPF/DKIM, set up IAM policy, use SES SMTP or API.

3) Payment Gateway

- **Best / Default (India): Razorpay (recommended)**
  - Why: widely used in India, solid dashboard, payouts (RazorpayX), documentation, 2% per transaction standard pricing.
  - Cost: ~2% + GST on most domestic transactions; enterprise pricing negotiable for large volumes.
  - Onboard notes: KYC (business), bank account, webhook endpoint, test keys then production keys.
- Cheapest alternative: PayU / Instamojo — sometimes cheaper for specific categories or promotional pricing; compare per-method fees and settlement times.

4) Cloud Hosting (API, frontend, background workers)

- **Best (enterprise): AWS (Mumbai region)**
  - **Why**: global availability zones, wide service portfolio (EC2, ECS, EKS, RDS, S3, CloudWatch), security & compliance. Good for hospital-grade data protection and backups. (Use VPC, private subnets, backups).
  - **Cost:** highly variable; small production infra usually ₹5k+ / month; grow with usage.

- **Cheapest (MVP): DigitalOcean / Linode**
    - **Why:** simpler pricing and lower baseline cost for small servers. Droplets from ₹400–₹800 / month (depending on spec).
    - **Pros:** easier to manage for small teams.
    - **Cons:** less rich feature set & compliance features vs AWS.

5) Database (managed)

- **Best:** AWS RDS (MySQL/Postgres) — managed backups, HA, automated snapshots.
- **Cheapest:** DigitalOcean Managed DB or self-hosted PostgreSQL on a droplet (lower cost but more ops)**.**

6) Object Storage (reports, scans)

- **Best:** AWS S3 — lifecycle policies, encryption, access control.
- **Cheapest (MVP):** DigitalOcean Spaces or Google Cloud Storage (compare promos).
- **Typical:** ₹1–₹2 / GB/month (variable).

7) Push Notifications / Mobile (free tier strong)

- **Best & Cheapest:** Firebase Cloud Messaging (FCM) — free, works well for Android/iOS web push.
- **Alternate:** OneSignal (free up to a point, paid for advanced segmentation).

8) Logging, Monitoring & Error Tracking

- **Best (enterprise):** Datadog (full APM + infra) — expensive, powerful**.**
- **Best (cost-effective): Sentry — excellent error tracking, free tier available; paid for retention & support. (Good starter choice.)**
- **Onboard note: integrate Sentry SDK in backend & frontend.**

9) Background jobs & queues

- **Best:** AWS SQS + Lambda / AWS Batch or RabbitMQ / Redis + BullMQ on managed infra.
- **Cheapest (MVP):** Redis (managed) + Bull / Sidekiq (self-hosted)

10) Backup & Disaster Recovery

- **Best:** AWS Backup (with cross-region replication to S3 Glacier).
- **Cheapest:** scheduled snapshots + S3 cold storage lifecycle rules.

11) PDF generation (invoices, prescriptions)

- **Cheapest (recommended):** server-side renderer (wkhtmltopdf, Puppeteer/Headless Chrome`) one-time cost, self-hosted.
- **Paid SaaS:** DocRaptor / PDFShift / PDFMonkey — simpler but ongoing cost.

12) Domain & SSL

- **Domain:** Namecheap / GoDaddy (cheap).
- **SSL:** Let's Encrypt (free) for TLS; paid EV/OV certs only if your hospital needs extended validation.

Recommended combos (quick)

- **MVP (lowest monthly):**
    - **SMS:** Fast2SMS or MSG91 (prepaid) — DLT done via aggregator.
    - **Email:** Amazon SES.
    - **Payments:** Razorpay.
    - **Hosting:** DigitalOcean droplet + Managed DB.
    - **Storage:** DigitalOcean Spaces.
    - **Push:** Firebase.
    - **Monitoring:** Sentry.
- **Enterprise (hospital-grade):**
    - **SMS:** Twilio (or Twilio + local aggregators for fallback).
    - **Email:** SendGrid or Amazon SES (SES cheaper).
    - **Payments:** Razorpay (2% + GST).
    - **Hosting:** AWS with multi-AZ, RDS, S3, CloudFront.
    - **Monitoring:** Datadog + Sentry.
    - **Backup:** AWS Backup + cross-region replication.

Estimated costs (India) — monthly ballpark (MVP vs Enterprise)

- **SMS (1,000 msgs/month):** MVP ~₹120 – ₹300; Enterprise (Twilio routes) ~₹800–₹2,000.
- **Email (5,000 txn/month):** SES < ₹200; SendGrid plan ≈ $20+/month.
- **Payments:** transaction cost 2% + GST on revenue (Razorpay).
- **Hosting + DB:** MVP (small DO droplet + managed DB): ₹800 – ₹3,000 / month. Enterprise (AWS small prod infra): ₹5,000 – ₹30,000+ / month depending on scale.
- **Storage:** ₹200 – ₹1,000 depending on GBs.
- **Monitoring & backup:** ₹500 – ₹5,000 depending on plan.

Total MVP monthly: roughly ₹5,000 – ₹10,000 (matches earlier rough estimate in your brief).
Total enterprise monthly: ₹15,000 – ₹60,000+ depending on scale & SLA.

Onboarding checklist (concrete steps developers / ops team can follow)

1. **Business setup & legal**
    - Ensure hospital has registered business entity (PAN, GST if applicable). Payment gateways require KYC.
2. **DLT & SMS**
    - Decide SMS provider (aggregator vs Twilio).
    - Register DLT principal entity (on any operator DLT portal).
    - Register Sender IDs (headers).
    - Register content templates (OTP, booking confirmation, reminders).
    - Test OTP sending and track delivery receipts.
3. **Payment gateway**
    - Create Razorpay business account, complete KYC, configure webhooks (payment/success/refund).
4. **Email**
    - Create SendGrid or SES account, verify domain, add SPF/DKIM, set bounce handling.
5. **Hosting & infra**
    - Provision staging & prod environments (separate VPCs, DBs).
    - Setup CI/CD (GitHub Actions or equivalent).
    - Setup monitoring & alerting (Sentry/Datadog + PagerDuty).

6. **Security**
    - HTTPS everywhere, strong JWT or OAuth2 flows, role based access controls.
    - Data encryption at rest + in transit.
    - Daily encrypted backups.
7. **Contracts & SLA**
    - Sign support SLA with SMS/email providers for escalation & credits.
    - Confirm payment gateway settlement SLA (T+1/T+2) with Razorpay.

Recommended SLA & contract items to request from vendors

(important for hospital uptime & legal)

- 99.9% uptime for messaging & payment APIs (or credits when not met)
- Delivery reporting and retry/fallback behavior for SMS (essential for OTP)
- Support response: P1 incidents within 1 hour (phone + ticket)
- Data residency: confirm where user data is stored (important if you need India-only storage)
- Logging & audit access for payments (for reconciliation)

Quick integration notes & code pointers (high level)

- **OTP flow**
    1. Generate OTP server-side (6 digits), store hashed with TTL (e.g., Redis 5 min).
    2. Call SMS provider API (use Verify API if Twilio) with registered template & sender ID.
    3. On delivery webhook, update OTP status (optional).
    4. On success verify OTP against hash.
- **Payment flow (Razorpay)**
    1. Create Order on server (Razorpay Orders API) → return order_id to client.
    2. Client opens Razorpay checkout; on success call backend to verify signature and capture.
    3. Use webhooks to reconcile payments asynchronously (recommended).
- **Email receipts (SES)**
    - Use verified domain, prebuilt template; send via SES API & subscribe to bounce/complaint SNS topics.

Legal & compliance highlights (must)

- DLT compliance for any bulk/transactional SMS in India.
- Consent management: store opt-ins/opt-outs for patients.
- Data retention & privacy: publish privacy policy and data retention schedule (medical records often have local legal retention requirements).
- Payment KYC & invoices: ensure GST invoices for payments & refunds.

Practical suggested purchase plan (step-by-step)

1. Register hospital business entity (if not already) — required for Razorpay & DLT.
2. Create developer accounts: (Razorpay sandbox, SES dev, Fast2SMS test)
3. Complete DLT principal entity registration (or ask aggregator to do it).
4. Buy initial SMS credits (Fast2SMS) or Twilio account + sender IDs.
5. Provision staging server (DigitalOcean droplet), set up SSL (Let's Encrypt), and test end-to-end flows (book → pay sandbox → send OTP → email).
6. Move to production (AWS) when traffic & compliance needs grow.

SLA / contract language to request from a paid vendor

Vendor will maintain at least 99.9% service availability measured monthly. Vendor will provide a dedicated escalation contact and respond to P1 incidents within one hour. For SMS: Vendor will provide delivery reports and message retry logic; vendor will provide assistance for DLT template registration and maintain delivery logs for at least 90 days.

Appendix — Useful reference links (for your ops/dev team)

- Twilio SMS pricing & docs.
- Fast2SMS pricing.
- MSG91 pricing & guides.
- SendGrid pricing.
- Amazon SES pricing & docs.
- Razorpay pricing & docs.
- DLT registration guides (TRAI / aggregator explainers).